

PART A: PROJECT OVERVIEW

1. Project Idea Definition

Project Topic: Library Management System

Our project focuses on designing a Library Management System (LMS) which helps libraries to organise and operate their resources in a more effective manner. The system is based on the ERD which we created to include essential objects which contain Books and Book Copies and Authors and Publishers and Members and Staff and Loans and Fines and Reservations. The database system uses these entities to create a structured database system which replaces the manual processes that most libraries currently use. The system maintains all library data in a single digital database which eliminates the need to use paper documents and physical files. The system enables library staff to execute their work tasks with higher speed and accuracy and improved control while providing students and library users with better access to educational resources.

The libraries throughout Kenya continue to use manual systems for record-keeping purposes which they utilize to handle their operational work. The library services face multiple problems because of these systems which impede their operational performance.

1. Poor Record Keeping

The majority of libraries use logbooks and notebooks and paper files to keep records about their books and library members. The records can become lost or damaged through various means because

humans make mistakes while they update the records which take time to search for information.
Staff

members face challenges in maintaining correct and well-organized records because the information exists across multiple storage locations.

2. Difficulty Tracking Borrowed Books

The absence of a digital system forces librarians to perform manual checks about who exists as the current book borrower and whether the book has returned. The procedure takes a long time to complete while it produces results that scientists find difficult to trust. The library loses track of its assets because members forget to return materials and books get misplaced while overdue items remain unmonitored.

3. No Automatic Fine Calculation

The process of calculating fines through manual methods leads to high rates of errors. The system produces problems because it creates inconsistent fine charges which staff members and members misinterpret while they charge incorrect fine amounts which results in lost revenue through unrecorded fines.

4. Poor Book Availability Management

The libraries face difficulties in obtaining precise information about their book collection status. The situation creates obstacles for determining total book inventory because it requires tracking current loans and overdue items and reserved copies.

5. Limited Staff Accountability

The manual record-keeping system allows various staff members to change records without any system that tracks their actions. The system creates challenges for tracking who issued books and who

made errors during the recording process and who did not return books.

6. Slow Service Delivery

The library needs to use manual systems because its existing systems cannot meet their requirement to

operate during peak times which happens during examination periods. The staff needs to complete multiple tasks which include searching files to find member information and checking book status and

tracking borrow history and determining fines. This process creates extended wait times which result in service interruptions for users.

B. Context in the Kenyan Environment

The local challenges which affect Kenyan libraries create the problems which libraries experience throughout the country.

1. Limited Budgets

School libraries and public institution libraries operate with budget constraints which limit their financial resources. The institutions face difficulties because they lack funds to purchase

contemporary library software and digital systems and computer hardware and staff development programs.

2. Large Number of Library Users

Libraries in Kenya serve multiple user groups which include students and teachers and researchers and community members. The process of handling such extensive user bases through manual methods

becomes unmanageable because it creates organizational challenges.

3. Rural Library Challenges

The rural community libraries and rural school libraries need to use handwritten registers because they lack reliable information and communication technology systems and skilled personnel and adequate financial support.

4. Lack of Trained Library Staff

Some organizations use teachers and administrative personnel to perform librarian duties because these individuals lack formal library management education. The process of handling records through

manual methods creates a higher probability of mistakes.

C. Connection to Sustainable Development Goal (SDG)

SDG 4: Quality Education

Our system supports SDG 4 which's about improving access to quality education and promoting lifelong learning.

1. Improving access to learning materials

Our system provides students and community members with easy access to necessary books.

2. Strengthening literacy and learning culture

Our system helps people develop a reading habit and research skills through better library operations which create an efficient learning environment.

3. Enhancing education management

The system enables libraries to store both records and reports which helps them decide which books to purchase and which resources to allocate.

4. Supporting learning

Our system enables libraries to deliver effective services to all age groups which supports their ongoing education and personal growth.

D. How the Database System Helps Solve the Problem

- 1) The system provides unified data storage through its centralized system which enables complete access to library information without any missing or conflicting records.
- 2) Our system automatically records when a book is borrowed, when it is due back and who borrowed it.
- 3) The system performs automatic fine calculations which guarantee that all calculated fines remain both accurate and equitable.
- 4) The system displays all book statuses by indicating which books are accessible and which ones are currently borrowed or have passed their return date.
- 5) The system enables smoother operations through its book and member search functions which help decrease customer waiting periods.
- 6) The system records every transaction according to its design which establishes an environment of trustworthy records that enable staff members to maintain accountable operations.
- 7) The system produces reports which assist libraries in developing their strategic plans and making operational decisions.